

Supplementary Information: Molecular SKALA in CP2K

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1 Implementation Scope

The implementation described in the main manuscript is deliberately molecular. It supports isolated GPW calculations with GTH pseudopotentials and isolated all-electron GAPW calculations with `POTENTIAL ALL`. In both cases, CP2K supplies the molecular geometry, Gaussian basis, spin-resolved AO density matrix, and communicator to GAUXC, and receives the XC energy, AO XC matrix, and the available nuclear derivatives. RKS and UKS calculations, CDFT checks, CDFT-CI checks, force checks, and molecular virial checks are included in this scope.

Cases that would require a different density contract are kept outside the present implementation. In particular, GAPW with GTH pseudopotentials and the native `GAPW_XC` correction are not enabled for SKALA, because the smooth, hard, soft, and compensation density terms cannot be replaced by a single external nonlocal AO-density evaluation without an additional derivation. Periodic Γ -point calculations, k-point sampling, analytical periodic stress, and higher XC response kernels are likewise left for future work.

2 Computational Protocol

All validation calculations use isolated molecular boundary conditions. The main-manuscript calculations use all-electron GAPW with `POTENTIAL ALL` and all-electron MOLOPT_UZH basis sets. The small-molecule validation table and the water-hexamer application use QZVPP-quality all-electron MOLOPT_UZH basis sets. The complementary pseudopotential checks reported below use GPW with PBE-optimized UZH molecularly optimized GTH basis sets and matching PBE GTH pseudopotentials.

For native CP2K PBE references in all-electron GAPW, `GAPW_ACCURATE_XCINT` is enabled. This option evaluates the augmented XC contribution with the hard all-electron and soft compensation one-center density terms explicitly included in the XC integration. It therefore makes the native CP2K PBE reference consistent with the all-electron density represented by the AO density matrix passed to GAUXC. Without this setting, one would compare the external all-electron AO-density path to a cheaper native GAPW XC approximation rather than to the most complete native all-electron reference.

The PBE-through-GAUXC validation runs use the same molecular geometry, basis, density-matrix spin convention, SCF thresholds, and numerical grids as the corresponding native CP2K PBE runs. The only intended difference is the source of the XC energy and AO XC matrix. SKALA calculations then reuse the same interface with the learned model selected inside GAUXC. Unless stated otherwise, diagnostic energy differences are reported in hartree, force errors in hartree/bohr, and molecular-virial errors in hartree.

3 CPU/GPU Timing Diagnostic

The CP2K–GAUXC–SKALA interface is suitable for a focused CPU/GPU timing comparison because the learned model evaluation remains inside the GAUXC/LibTorch layer. The validation runs in the main manuscript use the host path to keep the AO density-matrix, force, and molecular-virial checks independent of device-specific effects. The timings below are therefore reported as a performance diagnostic of the same interface, not as additional production-quality energy benchmarks.

The measurements were run on the Spark workstation with an NVIDIA GB10 device. They use isolated molecular GPW+GTH water clusters, the SKALA 1.1 checkpoint through GAUXC, one SCF Kohn–Sham matrix build from an atomic guess, $E_{\text{cut}} = 150$ Ry, $E_{\text{rel}} = 30$ Ry, and the GAUXC FINE/ROBUST molecular quadrature. This deliberately lightweight grid was chosen because the SUPERFINE/UNPRUNED one-water CUDA run already reached the memory limit of the 120 GB GPU. Thus the table is a hardware-path diagnostic; the grid-converged validation and water-hexamer energetics remain the main manuscript benchmarks.

The useful quantities are the total wall time, the CP2K timing-tree total, the KS-matrix construction time, and the speedups

$$S_{\text{XC}} = \frac{t_{\text{XC}}^{\text{CPU}}}{t_{\text{XC}}^{\text{GPU}}}, \quad S_{\text{SCF}} = \frac{t_{\text{SCF}}^{\text{CPU}}}{t_{\text{SCF}}^{\text{GPU}}}. \quad (\text{S1})$$

Here S_{XC} is approximated by the `qs_ks_build_kohn_sham_matrix` timing-tree entry, which contains the GauXC/SKALA energy-and-potential evaluation together with the surrounding KS-matrix insertion work. The end-to-end wall-time speedup is more conservative because it includes process start-up, CUDA context initialization, model loading, and the remaining CP2K setup.

Table S1: Spark CPU/GPU SKALA-through-GAUXC timing diagnostic for isolated $(\text{H}_2\text{O})_n$ clusters. Times are in seconds. The CPU reference uses host GAUXC/LibTorch; the GPU run uses device load-balancer and integrator execution.

n	Atoms	$t_{\text{wall}}^{\text{CPU}}$	$t_{\text{wall}}^{\text{GPU}}$	S_{wall}	$t_{\text{KS}}^{\text{CPU}}$	$t_{\text{KS}}^{\text{GPU}}$	S_{KS}	ΔE (Ha)
1	3	4.37	7.74	0.56	3.47	1.77	1.96	1.07×10^{-8}
2	6	8.07	8.05	1.00	7.15	1.96	3.64	3.47×10^{-8}
4	12	19.77	9.72	2.03	18.76	3.51	5.35	8.14×10^{-8}
8	24	75.64	25.62	2.95	70.86	12.45	5.69	1.55×10^{-7}

The timing trend is consistent with the expected division of labor. For one water molecule, fixed launch and model-initialization costs dominate the wall time and the GPU is only faster in the KS-matrix timing-tree entry. From four water molecules onward, the end-to-end timing also benefits from device execution, and for $(\text{H}_2\text{O})_8$ the KS-matrix entry is accelerated by $5.7\times$ while the total wall time is accelerated by $3.0\times$. The CPU/GPU energies agree to better than 2×10^{-7} Ha for all tested clusters, confirming that the timing diagnostic exercises the same molecular SKALA energy path on both backends.

4 Meta-GGA Grid Sensitivity

Table S2 gives the closed-shell H_2O kinetic-energy-density diagnostic at the original $E_{\text{cut}} = 600$ Ry and $E_{\text{rel}} = 60$ Ry grid settings. The main manuscript reports the corresponding 1200/80 Ry rerun, for which the GPW+GTH energy mismatches are smaller while the force and molecular-virial finite-difference checks remain at the 10^{-7} hartree/bohr and hartree level. This behavior is consistent with the generally slower grid convergence of meta-GGA functionals, where the Kohn–Sham kinetic-energy density is a functional ingredient in addition to the density and its gradient. The comparison illustrates that the residual TPSS and r^2SCAN energy differences are dominated by grid sensitivity in the kinetic-energy-density ingredients rather than by a failure of the AO matrix or derivative path.

Table S2: Closed-shell H_2O meta-GGA kinetic-energy-density diagnostic at $E_{\text{cut}} = 600$ Ry and $E_{\text{rel}} = 60$ Ry. $\Delta E = E^{\text{GAUXC}} - E^{\text{CP2K}}$ is given in hartree. Force errors are absolute analytical-minus-finite-difference differences in hartree/bohr; molecular virial errors are in hartree.

Method	Functional	ΔE	$ \Delta F _{\text{CP2K}}$	$ \Delta F _{\text{GAUXC}}$	$ \Delta W _{\text{CP2K}}$	$ \Delta W _{\text{GAUXC}}$	$ \Delta W_{\text{xc}} _{\text{GAUXC}}$
GPW+GTH	TPSS	-7.019×10^{-4}	1.29×10^{-5}	4.84×10^{-7}	7.38×10^{-5}	4.44×10^{-7}	1.05×10^{-8}
GPW+GTH	r^2SCAN	-1.805×10^{-4}	1.23×10^{-5}	4.88×10^{-7}	5.46×10^{-5}	1.78×10^{-7}	9.92×10^{-9}
GAPW all-electron	TPSS	-5.222×10^{-4}	5.14×10^{-7}	5.19×10^{-7}	3.35×10^{-7}	3.21×10^{-7}	1.15×10^{-8}
GAPW all-electron	r^2SCAN	$+9.28 \times 10^{-5}$	5.30×10^{-7}	5.22×10^{-7}	4.13×10^{-8}	3.04×10^{-8}	1.18×10^{-8}

5 Finite-Difference Checks

Forces are validated against central finite differences of the total energy,

$$F_{A,i}^{\text{FD}} = -\frac{E(R_{A,i} + \delta) - E(R_{A,i} - \delta)}{2\delta}. \quad (\text{S2})$$

The displacement δ is chosen small enough to probe the local derivative but large enough to avoid SCF and quadrature noise. The reported values in the main manuscript are absolute differences between the analytical force component printed by CP2K and Eq. (S1).

The molecular virial diagnostic is not a periodic cell stress. It is obtained from an affine scaling of all molecular coordinates around a fixed center,

$$\mathbf{R}_A(\lambda) = \mathbf{R}_c + (1 + \lambda) (\mathbf{R}_A - \mathbf{R}_c), \quad (\text{S3})$$

and the finite-difference scalar

$$W^{\text{FD}} = - \left. \frac{\partial E[\{\mathbf{R}_A(\lambda)\}]}{\partial \lambda} \right|_{\lambda=0} \simeq - \frac{E(+\lambda) - E(-\lambda)}{2\lambda}. \quad (\text{S4})$$

This check is useful for molecular forces and coordinate scaling, but it omits the additional cell, grid, image, reciprocal-space, and Pulay derivatives needed for a true periodic stress tensor.

6 Water-Hexamer Benchmark Details

The water-hexamer application uses structures and reference binding energies from the updated water-cluster subset of BEGDB. The selected neutral hexamer isomers are prism, cage, book 1, book 2, bag, cyclic chair, cyclic boat 1, and cyclic boat 2. The reference values in the main manuscript are the BEGDB CCSD(T)/CBS no-counterpoise binding energies converted to relative energies without zero-point corrections.

All production calculations in this section use all-electron GAPW, `POTENTIAL ALL`, QZVPP-quality MOLOPT_UZH basis sets, and `GAPW_ACCURATE_XCINT`. The latter option is retained for the native PBE reference so that the one-center augmented XC terms include the hard all-electron and soft compensation density contributions. This is the appropriate native CP2K comparison for the all-electron AO-density matrix evaluated by GAUXC.

The PBE-D3(BJ) values are direct native CP2K calculations with the Becke–Johnson damped D3 correction. The SKALA-D3(BJ) values add a separate B3LYP5-D3(BJ) binding-energy correction to the SKALA XC binding energies, following the public SKALA 1.1 model metadata. This additive treatment is used because direct combinations of `XC_FUNCTIONAL/GAUXC` and `VDW_POTENTIAL` are deliberately rejected in the present CP2K interface until the combined energy, force, and stress contract has been validated.

Each method is referenced to its own lowest-energy isomer,

$$\Delta E_i = E_i - \min_j E_j. \quad (\text{S5})$$

This avoids mixing absolute-energy offsets and focuses the comparison on the relative ordering within a fixed Hamiltonian and density representation. The structures, input files, raw outputs, and processed binding and relative energies are included in the accompanying data package.

Table S3 summarizes the auxiliary water-hexamer data sets. The QZVPP all-electron GAPW protocol is the main-manuscript data set. The TZVPP all-electron GAPW protocol probes basis-set sensitivity, and the GPW+GTH protocol probes the same workflow in the pseudopotential representation. The maximum PBE mismatch is the largest absolute difference between GAUXC-PBE and native CP2K PBE relative energies for the eight isomers.

Table S3: Water-hexamer protocol sensitivity. All entries except the maximum PBE mismatch are mean unsigned relative-energy errors in kcal mol^{−1} with respect to the BEGDB reference.

Protocol	Max. PBE mismatch	GAUXC-PBE	PBE	PBE-D3(BJ)	SKALA XC	SKALA-D3(BJ)
GPW+GTH PBE	0.125	6.11	6.08	6.94	5.62	6.97
GAPW AE TZVPP	0.008	5.83	5.83	6.66	5.17	6.41
GAPW AE QZVPP	0.009	6.11	6.10	6.97	5.44	6.79

Table S4 gives the complementary GPW+GTH pseudopotential check for the same water-hexamer workflow. The calculations use PBE-optimized molecular GTH basis sets and PBE GTH pseudopotentials throughout. They are not used as the primary benchmark in the main manuscript, but they document that the same CP2K–GAUXC interface behaves consistently in the pseudopotential representation.

Table S5 gives the all-electron GAPW TZVPP basis-set sensitivity check. The headline benchmark in the main manuscript uses QZVPP-quality MOLOPT_UZH basis sets; the present auxiliary table keeps the smaller TZVPP basis to document basis-set sensitivity. The same qualitative pattern is obtained: GAUXC-PBE and native PBE agree closely, SKALA XC reduces the mean unsigned relative-energy error compared with PBE, and the additive D3(BJ) correction does not uniformly improve the water-hexamer ordering.

Table S4: Complementary GPW+GTH water-hexamer relative binding energies with PBE GTH basis sets and pseudopotentials. All values are in kcal mol⁻¹, relative to the lowest isomer within the respective method.

Isomer	Ref.	GAU XC-PBE	PBE	PBE-D3(BJ)	SKALA XC	SKALA-D3(BJ)
Prism	0.00	1.18	1.11	0.90	0.92	0.61
Cage	0.21	0.00	0.00	0.00	0.00	0.00
Book 1	0.63	10.30	10.29	11.14	10.10	11.41
Book 2	1.00	5.45	5.45	6.21	5.24	6.45
Cyclic chair	1.54	15.55	15.42	17.11	14.60	17.20
Bag	1.55	2.21	2.20	2.78	2.10	3.06
Cyclic boat 1	2.57	8.91	8.84	10.47	7.92	10.45
Cyclic boat 2	2.63	14.97	15.03	16.65	13.77	16.28
MUE	–	6.11	6.08	6.94	5.62	6.97

Table S5: Auxiliary all-electron GAPW TZVPP water-hexamer relative binding energies for basis-set sensitivity. All values are in kcal mol⁻¹, relative to the lowest isomer within the respective method.

Isomer	Ref.	GAU XC-PBE	PBE	PBE-D3(BJ)	SKALA XC	SKALA-D3(BJ)
Prism	0.00	0.78	0.77	0.56	0.64	0.33
Cage	0.21	0.00	0.00	0.00	0.00	0.00
Book 1	0.63	9.88	9.88	10.72	9.33	10.64
Book 2	1.00	4.72	4.72	5.48	4.29	5.50
Cyclic chair	1.54	15.64	15.64	17.33	14.23	16.83
Bag	1.55	1.41	1.40	1.99	1.09	2.05
Cyclic boat 1	2.57	8.76	8.76	10.39	7.35	9.87
Cyclic boat 2	2.63	14.86	14.86	16.48	13.25	15.76
MUE	–	5.83	5.83	6.66	5.17	6.41

7 Data Package

The data package accompanying the manuscript contains the molecular structures, CP2K inputs, selected raw outputs, processed tables, and short extraction scripts needed to reproduce the numerical tables in the manuscript. The final archive also contains checksums and machine-readable metadata. Diagnostic development files that are not required for reproducing the published tables are kept out of the Supplementary Information and, where retained for provenance, are clearly separated in the data package.